Context Integrity in LLM Agents

Two research communities study the same failure. Neither tests the other's work.
We measured what that costs.

Nyamtulla · Advisor meeting · 7 October 2026

An agent believes everything in its context window

An LLM agent reads one block of text and then acts. We call this block the context window.

The block holds the user's question. It also holds a page a tool returned, a document a search step found, and a note the agent stored last week.

The model cannot tell these apart. Everything in the block looks the same to it.

So any text that reaches the block can change what the agent does. The text does not need to come from the user. It only needs to arrive.

This is not about training data. It is about what enters the context while the agent runs.

Two communities, two vocabularies, one failure

	THE SECURITY SIDE	THE MACHINE LEARNING SIDE
What corrupts the context	an attack (182)	an incidental mechanism (55)
What the field builds against it	a defense (263)	a fix (266)
Who put it there	a person, on purpose	nobody
Examples	indirect prompt injection, poisoned retrieval	lost in the middle, knowledge conflict

— We keep both sets of words, because the two sides are not doing the same thing. A fix was never built to stop an adversary.

— A tool returns a page with a hidden order. A tool returns a page that is ten times too long. Both arrive by the same route and end in the same window. One has an author.

— Our question: does that difference justify two separate sets of protection?

Security side: Greshake et al., AISec@CCS 2023, arXiv:2302.12173 · ML side: Liu et al., TACL 2023, arXiv:2307.03172; Shi et al., ICML 2023, arXiv:2302.00093 · Knowledge conflict: Xu et al., EMNLP 2024, arXiv:2403.08319

We describe every case with three questions

TAG	THE QUESTION IT ANSWERS	VALUES
Channel	How did the text get in? Which route did it travel?	9
Intent	Did a person put it there on purpose, or not?	3
Consequence	What went wrong once the text was in the window?	6

— **9 channels x 3 intents x 6 consequences = 162 combinations.**

— A combination with no paper in it is a question the field has never asked. 98 of the 162 are empty.

— The framework tells us what is known, and it tells us where to look next.

CHANNEL

Nine routes carry text into the window

ROUTE	WHAT ARRIVES THIS WAY	NAMED ATTACKS
tool-output	a web page, an email, a file the agent opened	75
direct-input	the prompt itself, or the system prompt above it	67
RAG	a document a search step pulled from a corpus	28
cross-modal	text inside an image, audio or video	18
memory	something the agent stored earlier and reads again	16
tool-metadata	the description of a tool, not its output	13
multi-agent	a message from another agent	10
supply-chain	a package or template inherited before the run	6
skill	a stored procedure the agent loads and follows	4

tool-output Greshake et al., AISec@CCS 2023 · direct-input Liu et al., 2023 · RAG Zou et al., USENIX Security 2024 · memory Chen et al., NeurIPS 2024 · cross-modal Wang et al., EMNLP 2025
tool-metadata Wang et al., AAAI 2026 · multi-agent Lee et al., ESORICS 2025 · supply-chain Qu et al., 2026 · skill Jia et al., 2026

CONSEQUENCE

Six ways the agent then fails

FAILURE	WHAT IT MEANS	NAMED ATTACKS
goal-hijack	the agent does another person's task; text in the window gives an order and it obeys	136
reasoning-corruption	every word is true, but the window is too long or too cluttered to use well	46
silent-corruption	the agent answers from false content; nobody gave it a new task	21
persistence-backdoor	the bad content is stored and fires on a later run	16
data-exfiltration	the agent sends out what it can read	14
resource-abuse	the agent burns time, or refuses to answer at all	4

The second row is the machine learning literature. The other five are the security literature. That is the split, in one column.

goal-hijack Greshake et al., AISec@CCS 2023 · reasoning-corruption Liu et al., TACL 2023; Shi et al., ICML 2023 · silent-corruption Xie et al., ICLR 2023 · persistence-backdoor Chen et al., NeurIPS 2024 · data-exfiltration Qi et al., ICLR 2024 · resource-abuse Xue et al., 2024

A map of every attack and every defense in the field

COUNT	WHAT IT IS
1,026 papers	read in full and coded, drawn from about 26,000 candidates
182 attacks	deliberate ways to corrupt a context window
55 mechanisms	the same corruption arriving by accident
263 defenses	built to stop an adversary
266 fixes	built to stop the accident
349 pairs	every case where a paper tested one defense against one attack

— No prior survey holds both literatures. The closest reviews 78 papers, 37 attacks and 41 defenses, and stays on the security side.

— That map is the instrument. Everything that follows is read off it.

One row for every attack and every defense

— The map is a table, not a picture. One row for each named attack, one row for each named defense, 771 rows in all.

— Every row carries the same three tags. We read them from the paper's full text, not its title, and they come from fixed lists, so two papers describing the same thing land in the same place.

FIELD	AN ATTACK ROW	A DEFENSE ROW
Name	corpus poisoning	ParamMute
Route in	RAG	RAG
What goes wrong	silent-corruption	silent-corruption
Caused by a person	yes	no - built for the accident
From	Zhong et al., EMNLP 2023	Huang et al., NeurIPS 2025

— Same route, same failure, so the two rows share a cell and the defense has a claim to answer that attack.

— Nobody had ever run them against each other. We did.

FINDING 1

The two communities do not read each other

MEASURE	VALUE
Security papers citing at least one machine learning paper	11.7 %
Machine learning papers citing at least one security paper	9.1 %
Tested pairs that stay inside one literature	98.0 %
Combinations in the framework nobody has studied	60.5 %

— **Nine of ten papers never cross. Seven pairs in the whole field do.**

— This is not a sample. We resolved every citation edge in the corpus.

— **So every claim that a defense is robust holds inside one threat model. Nobody has checked the other one.**

FINDING 2

Causes that nothing has ever been run against

SIDE	CAUSE	NEVER ANSWERED
security	attack (182 named)	104
machine learning	incidental mechanism (55 named)	15

— Each paper proves its attack works, against a system with no defense switched on.

— What those papers compare against is other attacks, not defenses:

ATTACK	ITS PAPER COMPARES AGAINST
HOUYI	three earlier prompt injection techniques
JudgeDeceiver	six manual prompt injection attacks
ASB: Plan-of-Thought backdoor	five injection variants and five other attacks
Virtual Prompt Injection	clean training data and one concurrent poisoning method

— Each paper asks whether its attack beats the last attack. Whether any published defense stops it is never asked.

HOUYI Liu et al., 2023 · JudgeDeceiver Shi et al., ACM CCS 2024 · ASB Zhang et al., ICLR 2024 · VPI Yan et al., NAACL 2023

FINDING 2, THE OTHER SIDE

320 protections never run against any cause

— 124 of 263 defenses and 194 of 266 fixes have no tested pair. Reading all 320 shows what each one answers.

WHAT WE FOUND	COUNT	MOSTLY FROM	WHAT IT MEANS
A fix for an accidental failure	126	ML side	it answers a real failure; it never claimed to stop an adversary
Names a cause, never tests it	99	both	the cause is on our map; the experiment is missing
Threat described in prose only	41	security	an adversary is named; no experiment follows
Tests an attack it invented	22	security	evaluated only against the authors' own construction
Names a benchmark, not a cause	11	security	reports a score, not a protection outcome
Could not tell	21	security	not enough extracted text to decide

— 126 of them answer an accidental failure and never set out to stop an adversary. 99 name a cause that is on our map and stop short of testing against it.

— Those 99 are the gap, and the lead we follow next.

Where an untested pair is waiting

Each row is one route and one failure. Deliberate attacks on one side, and on the other the fixes built for that same failure arriving by accident.

Route in	What goes wrong	Attacks	Fixes	Ever tested against each other
direct-input	reasoning-corruption	4	79	**0**
RAG	silent-corruption	3	42	**0**
tool-output	goal-hijack	61	1	**0**
RAG	reasoning-corruption	2	17	**0**
direct-input	goal-hijack	27	1	**0**
direct-input	silent-corruption	1	11	**0**
memory	persistence-backdoor	7	1	**0**

Every row is zero. Not one of these pairs has been tested by anybody.

tool-output and goal-hijack alone holds 169 tested pairs, every one a defense against an attack or a fix against an accident. None crosses.

The RAG row is 3 attacks against 42 fixes. 126 possible tests. None run.

Released code is not the same as working code

We only report a number if we can run the authors' own code. So the first question is which code runs at all.

ATTACKS WE REBUILT	WHAT IT DOES TO THE ANSWER
corpus poisoning	removes the answer from the retrieved set
PoisonedRAG	inserts a false answer
BadRAG DoS	makes the agent refuse
BadRAG Selective-Fact	changes only the tone of a correct answer

DEFENSE CODEBASES WE OPENED	OUTCOME
4 of 9	run, and give a usable measurement
1 of 9	runs, but cannot reproduce its own published benefit
4 of 9	do not run: missing data, a missing prompt file, a checkpoint with no tokenizer

Nine defense codebases attempted. Four produced a measurement.

This sets the size of what the work can cover, and it is a finding in its own right.

The map tells us which pairs can even be run

A defense and an attack belong together when they sit in the same place on the map: the same route in, and the same thing going wrong.

If a defense was built for a failure that happens by accident, and an attack causes the same failure on purpose through the same route, then the defense has a claim to answer. Nobody has made it answer.

THE CELL WE STARTED IN

Route in	a document returned by a search step
What goes wrong	the agent answers from false content and nothing signals it
On the attack side	3 named attacks, all deliberate
On the fix side	42 named fixes, all built for the accidental version
Tested against each other	none of them, by anyone

The map picked the first experiments, not us. Three of the four defenses we ran sit in that cell. The fourth answers the same failure arriving by a different route.

We then widened the attack set along the same route. Two more attacks also poison a retrieved document, and testing those as well is how we checked that a result was about the defense and not about one attack.

The first measurements anyone has taken in that cell

WHAT WE RAN	DETAIL
4 attacks	rebuilt from released code and papers
5 defenses	counting two versions of one of them, all from released code
20 pairs	defense against attack, on three models, none of them reported anywhere
113 conditions	each pair run at several attack strengths, not one

Two rules decide whether we are allowed to claim anything.

A defense must first show its own benefit on its own task. If it cannot, we report nothing from it.

We fix in advance the smallest change the sample can detect. Below that we say nothing, in either direction.

Scoring is the authors' own code, unchanged.

THE RESULT

Does the defense stop the attack?

OUTCOME	WHAT IT MEANS	PAIRS
It stops the attack	the attack succeeds less often with the defense switched on	5
It does nothing	no readable difference either way	7
It makes the attack worse	the attack succeeds MORE often with the defense switched on	8

out of 20 pairs measured

A defense helped in 5 cases out of 20. In 8 it made the attack work better than no defense at all.

In 5 of those 8 the attack does almost nothing to a model with no defense at all. The defense is what makes it work.

That is the finding. A defense built for one kind of failure is not neutral against the other kind. It can be the reason the attack succeeds.

The same defense, the same attack, two models

We repeated pairs on a second model. The answer did not repeat with them.

DEFENSE AND ATTACK	ON LLAMA-3	ON MISTRAL	WHAT MOVED
Context-reliance vs tone steering	+95 pts worse	+20 pts worse	the size of the harm
Aggregation vs forced refusal	+85 pts worse	no readable effect	whether it happens at all
Context-reliance vs false answers	+9 pts worse	11 pts BETTER	which way it points

In the last row the same defense stops the attack on one model and helps it on the other. Not a smaller effect. The opposite effect.

So a result from one model is one measurement, not a property of the defense. Almost every paper in our map reports one model.

WHY IT HAPPENS

A cheap test predicts it. Two guesses failed.

An aggregating defense reads each passage on its own. So we measured what each model does when shown one poisoned passage alone, and compared that with what the defense then did overall.

	ACROSS FIVE MODELS
One-passage behaviour predicts the defended outcome	correlation 0.9999
The attack with no defense at all	0.0%, every model, every time

Two explanations we tested and had to drop

WE EXPECTED	WHAT HAPPENED
the attack needs a cautious model	the most safety-trained model we have never falls for it
the attack needs an obedient model	a model that obeys 70% of planted orders still never falls for it

What is left is narrower and more useful. The attack needs a model that lets a retrieved document decide whether it answers at all.

That is one cheap test away, and it is not the same thing as being safe. The safest model here is the one the attack cannot touch.

What is finished and what is still running

DONE

1 cell	worked through: the retrieved document, answering from false content
20 pairs	measured, none of them previously tested
3 models	including controlled repeats of the same pair on a second model
5 models	swept to find what makes one model break where another does not

RUNNING NOW

5 models	the second version of the aggregating defense, across the same five

Left to do

Measure directly whether a model lets a document decide if it answers — the property the sweep left pointing at, with no measurement yet attached.

One more attack in this cell: a third poisoning method with released code.

An aggregation defense from a different research group, to check the rule is not one laboratory's habit.

Then the next cell on the map. The largest untested pool is a different route in, with four attacks and seventy-nine defenses facing each other and nothing between them.

99 defense papers name an attack on our map and never test against it. Those are the next candidates after that.

The cell we finished took four attacks and five defenses. The map holds 10 more cells with both sides present and nothing tested between them.